

final-project-203a

```
library(tidyverse)
library(janitor)
library(caret)
library(nnet)
library(randomForest)
library(skimr)
library(tidymodels)
library(GGally)

tidymodels::tidymodels_prefer()
set.seed(20251212)

# Read data
fetal_raw <- read_csv("fetal_health.csv")

# Clean column names: 'baseline value' -> baseline_value, etc.
fetal <- fetal_raw %>%
  clean_names() %>%
  mutate(
    fetal_health = factor(
      fetal_health,
      levels = c(1, 2, 3),
      labels = c("Normal", "Suspect", "Pathological")
    )
  )

glimpse(fetal)
```

```
## Rows: 2,126
## Columns: 22
## $ baseline_value      <dbl> 120, 132, 133, ~
## $ accelerations       <dbl> 0.000, 0.006, 0~
## $ fetal_movement      <dbl> 0.000, 0.000, 0~
## $ uterine_contractions <dbl> 0.000, 0.006, 0~
## $ light_decelerations <dbl> 0.000, 0.003, 0~
## $ severe_decelerations <dbl> 0, 0, 0, 0, 0, ~
## $ prolonged_decelerations <dbl> 0.000, 0.000, 0~
## $ abnormal_short_term_variability <dbl> 73, 17, 16, 16,~
## $ mean_value_of_short_term_variability <dbl> 0.5, 2.1, 2.1, ~
## $ percentage_of_time_with_abnormal_long_term_variability <dbl> 43, 0, 0, 0, 0,~
## $ mean_value_of_long_term_variability <dbl> 2.4, 10.4, 13.4~
## $ histogram_width     <dbl> 64, 130, 130, 1~
## $ histogram_min       <dbl> 62, 68, 68, 53,~
## $ histogram_max       <dbl> 126, 198, 198, ~
```

```
## $ histogram_number_of_peaks      <dbl> 2, 6, 5, 11, 9, ~
## $ histogram_number_of_zeroes     <dbl> 0, 1, 1, 0, 0, ~
## $ histogram_mode                  <dbl> 120, 141, 141, ~
## $ histogram_mean                  <dbl> 137, 136, 135, ~
## $ histogram_median                <dbl> 121, 140, 138, ~
## $ histogram_variance              <dbl> 73, 12, 13, 13, ~
## $ histogram_tendency              <dbl> 1, 0, 0, 1, 1, ~
## $ fetal_health                    <fct> Suspect, Normal~
```

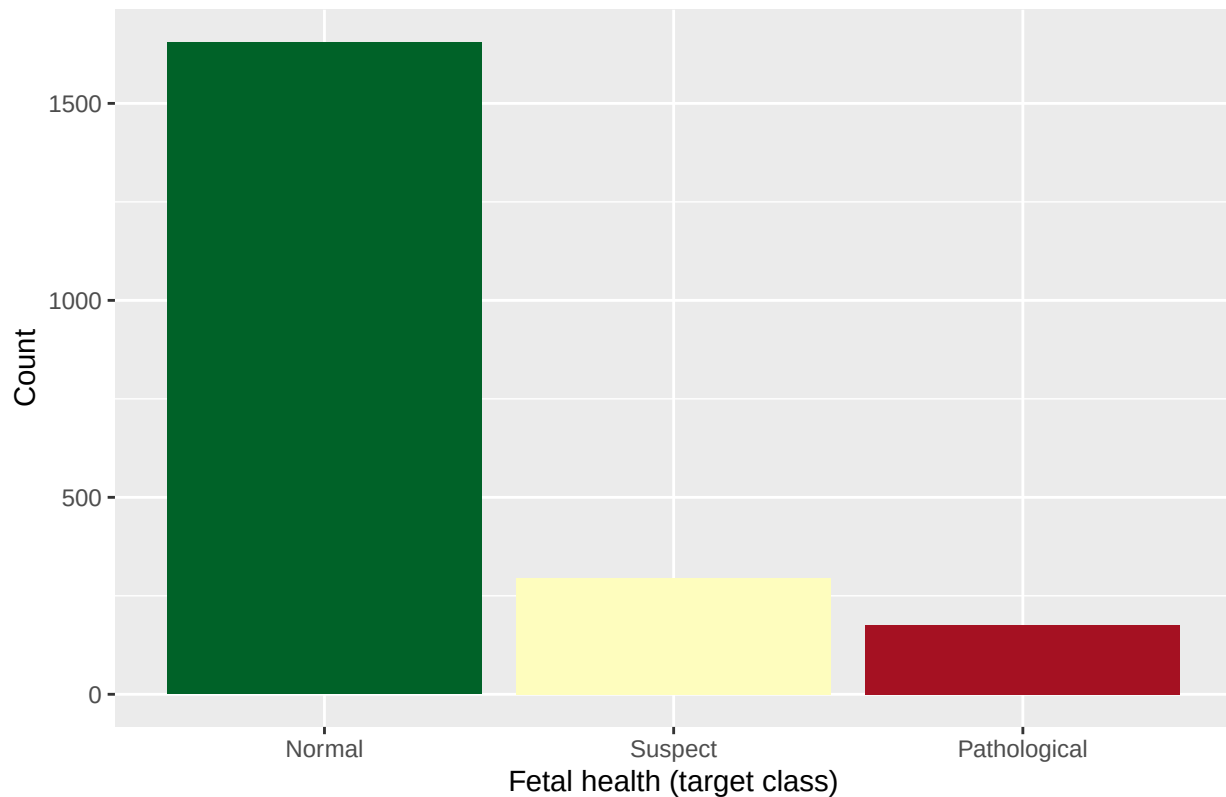
EDA

```
# outcome distribution
fetal %>%
  count(fetal_health) %>%
  mutate(prop = n / sum(n))
```

```
## # A tibble: 3 x 3
##   fetal_health     n   prop
##   <fct>         <int> <dbl>
## 1 Normal         1655 0.778
## 2 Suspect         295 0.139
## 3 Pathological   176 0.0828
```

```
fetal %>%
  ggplot(aes(x = fetal_health, fill = fetal_health)) +
  geom_bar() +
  scale_fill_manual(
    values = rev(hcl.colors(3, "RdYlGn")),
    guide = "none"
  ) +
  labs(
    x = "Fetal health (target class)",
    y = "Count",
    title = "Class distribution: Normal / Suspect / Pathological"
  )
```

Class distribution: Normal / Suspect / Pathological



```
# summary of predictors
fetal %>%
  select(-fetal_health) %>%
  summary()
```

```
## baseline_value accelerations fetal_movement uterine_contractions
## Min. :106.0 Min. :0.000000 Min. :0.000000 Min. :0.000000
## 1st Qu.:126.0 1st Qu.:0.000000 1st Qu.:0.000000 1st Qu.:0.002000
## Median :133.0 Median :0.002000 Median :0.000000 Median :0.004000
## Mean :133.3 Mean :0.003178 Mean :0.009481 Mean :0.004366
## 3rd Qu.:140.0 3rd Qu.:0.006000 3rd Qu.:0.003000 3rd Qu.:0.007000
## Max. :160.0 Max. :0.019000 Max. :0.481000 Max. :0.015000
## light_decelerations severe_decelerations prolonged_decelerations
## Min. :0.000000 Min. :0.000e+00 Min. :0.0000000
## 1st Qu.:0.000000 1st Qu.:0.000e+00 1st Qu.:0.0000000
## Median :0.000000 Median :0.000e+00 Median :0.0000000
## Mean :0.001889 Mean :3.293e-06 Mean :0.0001585
## 3rd Qu.:0.003000 3rd Qu.:0.000e+00 3rd Qu.:0.0000000
## Max. :0.015000 Max. :1.000e-03 Max. :0.0050000
## abnormal_short_term_variability mean_value_of_short_term_variability
## Min. :12.00 Min. :0.200
## 1st Qu.:32.00 1st Qu.:0.700
## Median :49.00 Median :1.200
## Mean :46.99 Mean :1.333
## 3rd Qu.:61.00 3rd Qu.:1.700
## Max. :87.00 Max. :7.000
```

```
## percentage_of_time_with_abnormal_long_term_variability
## Min. : 0.000
## 1st Qu.: 0.000
## Median : 0.000
## Mean : 9.847
## 3rd Qu.:11.000
## Max. :91.000
## mean_value_of_long_term_variability histogram_width histogram_min
## Min. : 0.000 Min. : 3.00 Min. : 50.00
## 1st Qu.: 4.600 1st Qu.: 37.00 1st Qu.: 67.00
## Median : 7.400 Median : 67.50 Median : 93.00
## Mean : 8.188 Mean : 70.45 Mean : 93.58
## 3rd Qu.:10.800 3rd Qu.:100.00 3rd Qu.:120.00
## Max. :50.700 Max. :180.00 Max. :159.00
## histogram_max histogram_number_of_peaks histogram_number_of_zeroes
## Min. :122 Min. : 0.000 Min. : 0.0000
## 1st Qu.:152 1st Qu.: 2.000 1st Qu.: 0.0000
## Median :162 Median : 3.000 Median : 0.0000
## Mean :164 Mean : 4.068 Mean : 0.3236
## 3rd Qu.:174 3rd Qu.: 6.000 3rd Qu.: 0.0000
## Max. :238 Max. :18.000 Max. :10.0000
## histogram_mode histogram_mean histogram_median histogram_variance
## Min. : 60.0 Min. : 73.0 Min. : 77.0 Min. : 0.00
## 1st Qu.:129.0 1st Qu.:125.0 1st Qu.:129.0 1st Qu.: 2.00
## Median :139.0 Median :136.0 Median :139.0 Median : 7.00
## Mean :137.5 Mean :134.6 Mean :138.1 Mean : 18.81
## 3rd Qu.:148.0 3rd Qu.:145.0 3rd Qu.:148.0 3rd Qu.: 24.00
## Max. :187.0 Max. :182.0 Max. :186.0 Max. :269.00
## histogram_tendency
## Min. : -1.0000
## 1st Qu.: 0.0000
## Median : 0.0000
## Mean : 0.3203
## 3rd Qu.: 1.0000
## Max. : 1.0000
```

```
library(tidyverse)
library(stringr)

# correlation matrix (predictors only)
X <- fetal %>% select(-fetal_health)
C <- cor(X, use = "pairwise.complete.obs")

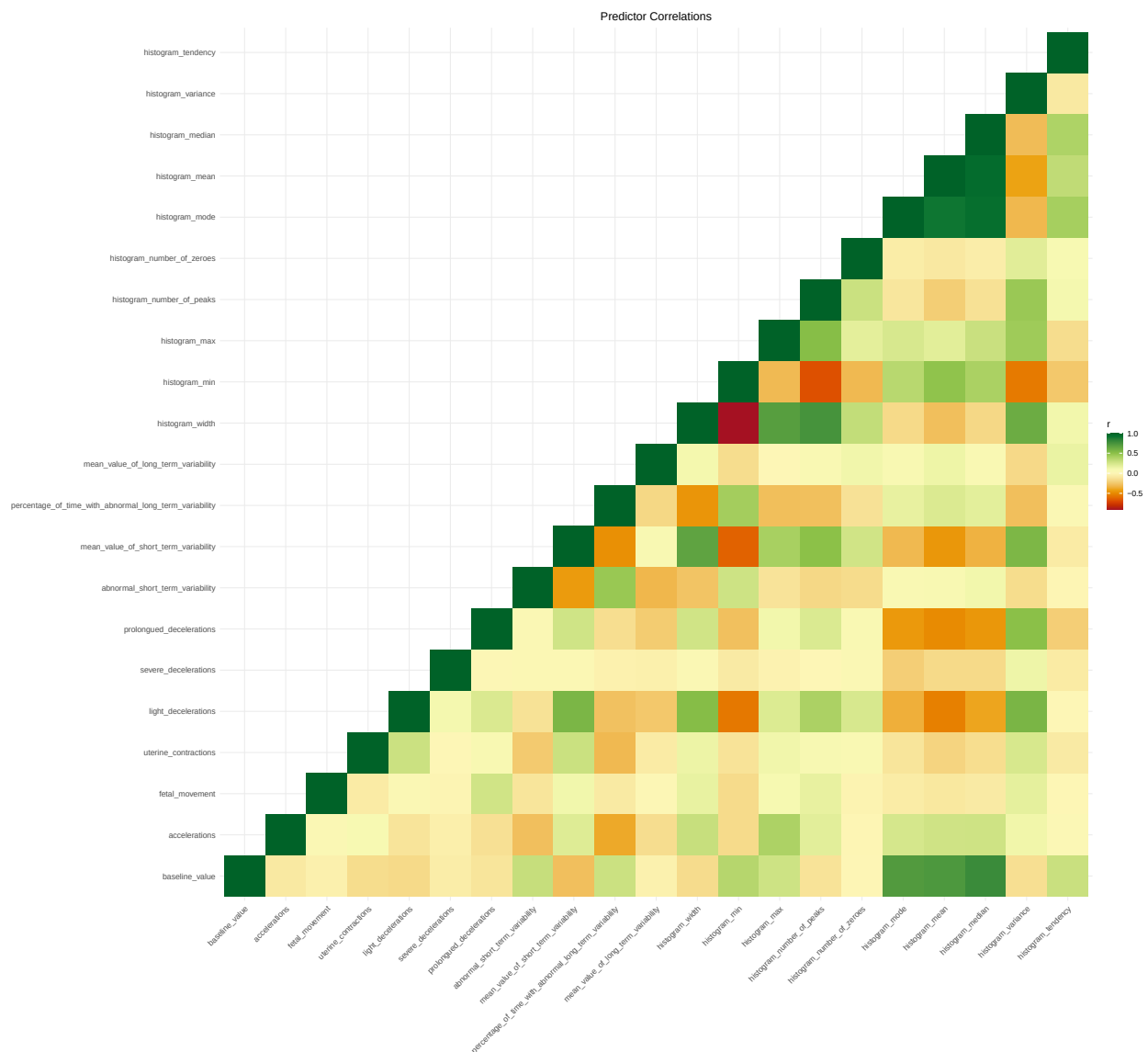
cor_long <- as.data.frame(C) %>%
  rownames_to_column("var1") %>%
  pivot_longer(-var1, names_to = "var2", values_to = "r")

cor_long <- cor_long %>%
  mutate(
    var1 = factor(var1, levels = colnames(C)),
    var2 = factor(var2, levels = colnames(C))
  ) %>%
  filter(as.integer(var1) <= as.integer(var2))
```

```

ggplot(cor_long, aes(var2, var1, fill = r)) +
  geom_tile() +
  coord_equal() +
  scale_fill_gradientn(colors = hcl.colors(20, "RdYlGn")) +
  scale_x_discrete(labels = function(x) str_wrap(x, width = 14)) +
  scale_y_discrete(labels = function(x) str_wrap(x, width = 14)) +
  labs(title = "Predictor Correlations", x = NULL, y = NULL, fill = "r") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 9),
    axis.text.y = element_text(size = 9)
  )

```



```
### Train/test split (stratified)
```

```
split_obj <- initial_split(fetal, prop = 0.80, strata = fetal_health)
train_dat <- training(split_obj)
test_dat <- testing(split_obj)
```

```
train_dat %>% count(fetal_health) %>% mutate(prop = n/sum(n))
```

```
## # A tibble: 3 x 3
##   fetal_health     n   prop
##   <fct>         <int> <dbl>
## 1 Normal         1319 0.776
## 2 Suspect         238 0.14
## 3 Pathological   143 0.0841
```

```
test_dat %>% count(fetal_health) %>% mutate(prop = n/sum(n))
```

```
## # A tibble: 3 x 3
##   fetal_health     n   prop
##   <fct>         <int> <dbl>
## 1 Normal         336 0.789
## 2 Suspect         57 0.134
## 3 Pathological    33 0.0775
```

Preprocessing recipe

```
fetal_recipe <- recipe(fetal_health ~ ., data = train_dat) %>%
  step_zv(all_predictors()) %>%
  step_normalize(all_numeric_predictors())
```

Fit models

Multinomial logistic regression

```
multinom_spec <- multinom_reg() %>%
  set_mode("classification") %>%
  set_engine("nnet", trace = FALSE)

multinom_wf <- workflow() %>%
  add_recipe(fetal_recipe) %>%
  add_model(multinom_spec)

multinom_fit <- fit(multinom_wf, data = train_dat)
```

Random forest

```

rf_spec <- rand_forest(trees = 1000) %>%
  set_mode("classification") %>%
  set_engine("ranger", importance = "impurity")

rf_wf <- workflow() %>%
  add_recipe(fetal_recipe) %>%
  add_model(rf_spec)

rf_fit <- fit(rf_wf, data = train_dat)

```

Evaluate models on test set

```

# Helper function: class-specific sensitivity from a confusion matrix
class_sensitivity <- function(cm) {
  m <- as.matrix(cm$table) # rows = truth, cols = prediction
  tibble(
    class = rownames(m),
    sensitivity = diag(m) / rowSums(m)
  )
}

# Multinomial logistic regression performance
multinom_pred <- predict(multinom_fit, test_dat, type = "class") %>%
  bind_cols(test_dat %>% select(fetal_health))

multinom_acc <- accuracy(multinom_pred, truth = fetal_health, estimate = .pred_class)

multinom_cm <- conf_mat(multinom_pred, truth = fetal_health, estimate = .pred_class)

multinom_sens <- class_sensitivity(multinom_cm) %>%
  filter(class %in% c("suspect", "pathological"))

multinom_acc

```

```

## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>       <dbl>
## 1 accuracy multiclass    0.908

```

```
multinom_sens
```

```

## # A tibble: 0 x 2
## # i 2 variables: class <chr>, sensitivity <dbl>

```

```

autoplot(multinom_cm, type = "heatmap") +
  labs(title = "Multinomial Logistic Regression: Confusion Matrix")

```

Multinomial Logistic Regression: Confusion Matrix

Prediction	Normal -	320	11	3
	Suspect -	15	38	1
	Pathological -	1	8	29
		Normal	Suspect	Pathological
		Truth		

```
# Random forest performance
rf_pred <- predict(rf_fit, test_dat, type = "class") %>%
  bind_cols(test_dat %>% select(fetal_health))

rf_acc <- accuracy(rf_pred, truth = fetal_health, estimate = .pred_class)

rf_cm <- conf_mat(rf_pred, truth = fetal_health, estimate = .pred_class)

rf_sens <- class_sensitivity(rf_cm) %>%
  filter(class %in% c("suspect", "pathological"))

rf_acc
```

```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>      <dbl>
## 1 accuracy multiclass    0.962
```

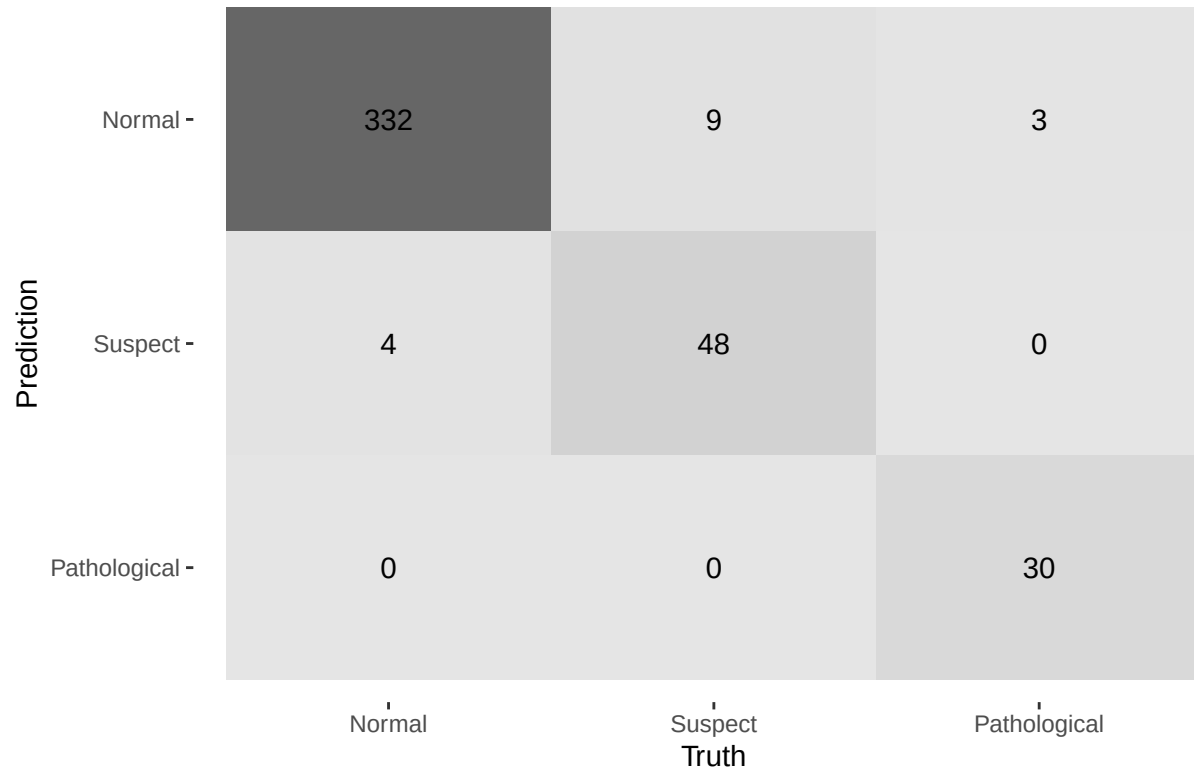
```
rf_sens
```

```
## # A tibble: 0 x 2
## # i 2 variables: class <chr>, sensitivity <dbl>
```



```
autoplot(rf_cm, type = "heatmap") +
  labs(title = "Random Forest: Confusion Matrix")
```

Random Forest: Confusion Matrix



Side-by-side model comparison

```
bind_rows(
  multinom_acc %>% mutate(model = "Multinomial logistic regression"),
  rf_acc %>% mutate(model = "Random forest")
) %>%
  select(model, .metric, .estimate)
```

```
## # A tibble: 2 x 3
##   model                .metric .estimate
##   <chr>                <chr>     <dbl>
## 1 Multinomial logistic regression accuracy    0.908
## 2 Random forest        accuracy    0.962
```

```
bind_rows(
  multinom_sens %>% mutate(model = "Multinomial logistic regression"),
  rf_sens %>% mutate(model = "Random forest")
) %>%
  select(model, class, sensitivity) %>%
  arrange(model, class)
```

```
## # A tibble: 0 x 3
## # i 3 variables: model <chr>, class <chr>, sensitivity <dbl>
```